Hengyi (Henry) Zhou

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Summary

I am a highly motivated dual degree student in mechanical engineering (SJTU) and computer engineering (Umich), interested in robotics (both algorithm and hardware). I plan to explore the interdisciplinary area and apply for a Ph.D. degree in the future.

Education

University of Michigan Ann Arbor, MI
Bachelor of Computer Engineering *Aug. 2025 – May 2027*

- Overall GPA: 4.00/4.00

Shanghai Jiao Tong University (SJTU) Shanghai, China
Bachelor of Mechanical Engineering *Aug. 2023 – May 2027*

- Overall GPA: 3.81/4.00

Experience

Research Group of Prof. Sean Huang Dec. 2025 – present
Undergraduate Research Intern *University of Michigan*

Investigate data-driven sensing and control for soft robotic systems, focusing on estimating contact forces and generating reliable motions when deformation depends on nonlinear equilibrium and actuation history. Combine physics-based models with neural network to address challenges in deformable objects manipulation and self-sensing soft structures, including collaborative work with UCLA.

Research Group of Prof. Ye Zhao Aug. 2026 – present
Undergraduate Research Intern *Georgia Tech, remote*

Study how reinforcement learning can support dexterous manipulation in medical applications, focusing on policy fine-tuning for precise and reliable robotic interaction.

Research Group of Prof. Chao Li Jan. 2025 – Jun. 2025
Undergraduate Research Intern *Shanghai Jiao Tong University*

Investigated low-latency retrieval-augmented generation (RAG) serving, focusing on reducing repeated computation and data movement through cache reuse and prefetching.

Publications

[1] **History-Aware Force Sensing and Tunable Stiffness in a Pneumatic Kresling Origami Cell**
Hengyi Zhou, Jiaqi Wang, Yilin Ma, Dezhong Tong, Xiaonan Huang
Under review at RoboSoft, 2027.

[2] **Self-Consistent Neural Control: Learning Trajectories for Implicit Equilibrium**
Hengyi Zhou, Haobo Fang, Xiaonan Huang, Dezhong Tong, M. Khalid Jawed
Under review at AAAI, 2027.
Spotlight, IROS Workshop BLPC, 2026.

[3] **Neural Control: Adjoint Learning Through Equilibrium Constraints**
Dezhong Tong, Jiawen Wang, ***Hengyi Zhou***, Yinlong Shen, Xiaonan Huang, M. Khalid Jawed
ICML, 2026. Accepted.

[4] **PCR: A Prefetch-Enhanced Cache Reuse System for Low-Latency RAG Serving**
Wenfeng Wang, Xiaofeng Hou, Peng Tang, ***Hengyi Zhou***, Jing Wang, Xinkai Wang, Chao Li, Minyi Guo
ACM Transactions on Architecture and Code Optimization (TACO). Accepted.

Projects

ME3500J Project | *SolidWorks, Python, Raspberry Pi* May 2025 – Aug. 2025

Designed and built a self-locking climber for cylindrical columns of varying diameters, using a dual-wheel stabilization mechanism to support stable climbing on rough surfaces. Demonstration: https://youtu.be/lMugocoG81E.

Teaching Experience

Grader for Intro to Signals and Systems Jan. 2026 – May. 2026
Umich

Teaching Assistant for Intro to Engineering May 2025 – Aug. 2025
Lecturer: Prof. Yanfeng Shen *SJTU*

Honors & Awards

Wang Chu Chien-Wen Research Award Apr. 2026
University of Michigan, 2 recipients each year

Jackson and Muriel Lum Scholarship & Fellowship Jul. 2025
University of Michigan & Shanghai Jiao Tong University, Top 2% in overall performance

Yu Liming Scholarship Nov. 2024
Shanghai Jiao Tong University, Top 5% in overall performance

Mathematical Contest in Modeling – Honorable Mention Jun. 2024
Consortium for Mathematics and Its Applications

Technical Skills

Languages: C/C++, Matlab, Python
Tools: Git, SolidWorks, Abaqus, WaveForm, Arduino, CubeIDE, Keil5, LabVIEW, Mathematica
Libraries: pandas, NumPy, PyTorch, JAX

Exchange

University of Navarra Jan. 2025 – Feb. 2025
Courses Taken: Espanol inicial: Comunicación y Cultura, Software Engineering *Navarra, Spain*

Korean University Dec. 2023 – Jan. 2024
Courses Taken: Introduction to Artificial Intelligence, Korean Entertainment *Seoul, South Korean*

Services

Minister of the JI Youth Volunteer Team May 2024 – May 2025
Shanghai Jiao Tong University *Shanghai, China*

Student Advisor in UMJI Advising Center Mar 2025 – present
Shanghai Jiao Tong University *Shanghai, China*